

$$\begin{array}{r}
 7 \overline{54 \ 91 \ 62} \ (741) \\
 \underline{49} \\
 144 \overline{5 \ 91} \\
 \underline{5 \ 76} \\
 1481 \overline{15 \ 62} \\
 \underline{14 \ 81} \\
 81
 \end{array}$$

∴ Required number to be subtracted = 81.

$$115. 0.000326 = \frac{326}{10^6}$$

$$\begin{array}{r}
 1 \overline{3 \ 26} \ (18) \\
 \underline{1} \\
 28 \overline{2 \ 26} \\
 \underline{2 \ 24} \\
 2
 \end{array}$$

∴ Required number to be subtracted = $\frac{2}{10^6} = 0.000002$.

$$\begin{array}{r}
 8 \overline{77 \ 00} \ (87) \\
 \underline{64} \\
 167 \overline{13 \ 00} \\
 \underline{11 \ 69} \\
 1 \ 31
 \end{array}$$

∴ Number to be added
 $= (88)^2 - 7700 = 7744 - 7700 = 44$.

$$\begin{array}{r}
 8 \overline{68 \ 06 \ 21} \ (824) \\
 \underline{64} \\
 162 \overline{4 \ 06} \\
 \underline{3 \ 24} \\
 1644 \overline{82 \ 21} \\
 \underline{65 \ 76} \\
 16 \ 45
 \end{array}$$

∴ Number to be added = $(825)^2 - 680621$
 $= 680625 - 680621 = 4$.

118. Greatest number of four digits is 9999.

$$\begin{array}{r}
 9 \overline{99 \ 99} \ (99) \\
 \underline{81} \\
 189 \overline{18 \ 99} \\
 \underline{17 \ 01} \\
 1 \ 98
 \end{array}$$

∴ Reqd. number = $(9999 - 198) = 9801$.

119. Least number of 4 digits is 1000.

$$\begin{array}{r}
 3 \overline{10 \ 00} \ (31) \\
 \underline{9} \\
 61 \overline{1 \ 00} \\
 \underline{61} \\
 39
 \end{array}$$

∴ $(31)^2 < 1000 < (32)^2$.

Hence, required number = $(32)^2 = 1024$.

120. Let the 18 consecutive natural numbers be $x, (x+1), (x+2), (x+3), \dots, (x+17)$. Then, $x + (x+1) + (x+2) + \dots + (x+17) = 18x + (1+2+3+\dots+17) = 18x + 153$.

Putting $x = 1, 2, 3, 4, \dots$ we find that the smallest value of x for which $(18x + 153)$ becomes a perfect square is $x = 4$.

∴ Required value = $18 \times 4 + 153 = 72 + 153 = 225$.

121. Given expression

$$\begin{aligned}
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2^2 - (\sqrt{2+\sqrt{2+\sqrt{3}}})^2} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{4 - (2+\sqrt{2+\sqrt{3}})} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2-\sqrt{2+\sqrt{3}}} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2^2 - (\sqrt{2+\sqrt{3}})^2} = \sqrt{2+\sqrt{3}} \cdot \sqrt{2-\sqrt{3}} \\
 &= \sqrt{2^2 - (\sqrt{3})^2} = \sqrt{4-3} = \sqrt{1} = 1.
 \end{aligned}$$

$$\begin{aligned}
 122. \frac{1}{(\sqrt{5}-\sqrt{3})} &= \frac{1}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} = \frac{(\sqrt{5}+\sqrt{3})}{(5-3)} \\
 &= \frac{(2.2361+1.7321)}{2} = \frac{3.9682}{2} = 1.9841.
 \end{aligned}$$

$$\begin{aligned}
 123. \text{ Given expression } &= \frac{1}{(\sqrt{9}-\sqrt{8})} \times \frac{(\sqrt{9}+\sqrt{8})}{(\sqrt{9}+\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} \\
 &\quad \times \frac{(\sqrt{8}+\sqrt{7})}{(\sqrt{8}+\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} \times \frac{(\sqrt{7}+\sqrt{6})}{(\sqrt{7}+\sqrt{6})} \\
 &\quad - \frac{1}{(\sqrt{6}-\sqrt{5})} \times \frac{(\sqrt{6}+\sqrt{5})}{(\sqrt{6}+\sqrt{5})} + \frac{1}{(\sqrt{5}-\sqrt{4})} \times \frac{(\sqrt{5}+\sqrt{4})}{(\sqrt{5}+\sqrt{4})} \\
 &= \frac{(\sqrt{9}+\sqrt{8})}{(9-8)} - \frac{(\sqrt{8}+\sqrt{7})}{(8-7)} + \frac{(\sqrt{7}+\sqrt{6})}{(7-6)} \\
 &\quad - \frac{(\sqrt{6}+\sqrt{5})}{(6-5)} + \frac{(\sqrt{5}+\sqrt{4})}{(5-4)} \\
 &= (\sqrt{9}+\sqrt{8}) - (\sqrt{8}+\sqrt{7}) + (\sqrt{7}+\sqrt{6}) - (\sqrt{6}+\sqrt{5}) \\
 &\quad + (\sqrt{5}+\sqrt{4}) = (\sqrt{9}+\sqrt{4}) = 3+2=5.
 \end{aligned}$$

$$\begin{aligned}
 124. \text{ Given exp. } &= \frac{1}{\sqrt{2}+\sqrt{1}} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} \\
 &\quad + \dots + \frac{1}{\sqrt{121}+\sqrt{120}} \\
 &= \frac{1}{\sqrt{2}+\sqrt{1}} \times \frac{\sqrt{2}-\sqrt{1}}{\sqrt{2}-\sqrt{1}} + \frac{1}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} \\
 &\quad + \frac{1}{\sqrt{4}+\sqrt{3}} \times \frac{\sqrt{4}-\sqrt{3}}{\sqrt{4}-\sqrt{3}} + \dots + \frac{1}{\sqrt{121}+\sqrt{120}} \times \frac{\sqrt{121}-\sqrt{120}}{\sqrt{121}-\sqrt{120}} \\
 &= \frac{\sqrt{2}-\sqrt{1}}{2-1} + \frac{\sqrt{3}-\sqrt{2}}{3-2} + \frac{\sqrt{4}-\sqrt{3}}{4-3} + \dots + \frac{\sqrt{121}-\sqrt{120}}{121-120} \\
 &= \sqrt{2}-\sqrt{1} + \sqrt{3}-\sqrt{2} + \sqrt{4}-\sqrt{3} + \dots + \sqrt{121}-\sqrt{120} \\
 &= -1 + \sqrt{121} = -1 + 11 = 10.
 \end{aligned}$$